

Intelligent model for Performance Improvement of Liquid Level Sensing System using Fiber Bragg Grating Sensor

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ABSTRACT

In this paper, a novel liquid-level sensing system is proposed to enhance the capacity of the sensing system and reduce the cost. The proposed sensing system can monitor the liquid level of several points at the same time. Additionally, for cost efficiency, the proposed system employs only one fiber Bragg grating (FBG) sensor in each container. Hence, an FBG sensor is connected to a properly designed float in order to detect liquid levels. When changing the liquid level inside the tank (water level container), the float position is changed and consequently, the axial strain is applied to the FBG. The axial strain causes a shift in the reflected wavelength of the FBG sensors. Hence, the water level of the tank is monitored depending on the wavelength shift. The wavelength shift of FBG leads to overlap or cross-talk between two FBG water level sensors and is very challenging to properly identify the water level of each sensor. To solve this overlap problem and to accurately predict the liquid level of each tank, we proposed a deep neural network (DNN) approach. The performance of the proposed DNN model is evaluated via two different scenarios. The result proves that the proposed DNN model accurately predicts the liquid level of each tank. Furthermore, when comparing the DNN model with the conventional machine learning schemes including random forest (RF) and support vector machines (SVM), DNN model exhibits the best performance.

1. Introduction

Accurately measuring liquid levels is a critical task in various industrial applications, such as chemical processes, oil and water reservoirs, fuel storage in transporting systems, and construction maintenance and monitoring [1]. To achieve this end, different types of sensors have been proposed. Electrical sensors are one of the most common types of sensors [2]. However, their safety is an issue when used in hostile conditions, particularly in explosive or combustible environments [3]. Mechanic sensors, like float sensors are low-cost devices that meet the safety requirement. However, they have low resolution, vulnerable to mechanical damage, and have expensive maintenance costs, leading to higher system weight and volume, and limiting their applicability [3]. In circumstances that needed contactless liquid level sensing, radio frequency radar-based sensors have been considered as a solution. Nevertheless, the polluted environment between the liquid and sensor can easily cause inaccuracy measurements [4].

Recently, Fiber optic sensors (FOSs) have been given priority over all other types of sensors [5–7]. The reason FOSs have a priority is that they have advantages such as their built-in characteristics, such as innate safety in terms of no power consumption, resistance, immunity to chemical corrosion, and electromagnetic interference. Furthermore, these sensors have received much attention due to their small size, large bandwidth, high accuracy and resolution, and especially high

multiplexing capacity. Various FOSs configurations for monitoring liquid levels [8–12] have recently been presented in recent years. One of the ordinary FOSs is fiber Bragg grating (FBG) sensors. Hence, using an array of FBG sensors and locating them at different levels of water to sense the pressure or the temperature of the water and determine the liquid level [8]. In [9], a fiber-optic liquid-level sensor based on FBG bending has been proposed to measure the water level. The FBG is embedded in a cantilever rod, and its elongation and contraction reflect a change in the liquid level. Also, in [10], etched chirped fiber Bragg grating (CFBG) is proposed as a liquid-level sensor. When the etched CFBG is submerged in liquid, its effective index changes, thus the phase and Bragg wavelength of the reflected spectrum will change. However, these schemes only concentrated on measuring the liquid level of a single spot or a single container as opposed to measuring multiple spots or containers at the same time. Still sensing the liquid level of different containers or spots at the same time is a challenge. Moreover, previous studies [11,12] use more than one sensor to measure the liquid level for one container or one spot. However, using numerous FBG sensors for only one container or spot is not cost-effective, particularly in industries that need to measure a large number of containers at the same time or at different points of a reservoir. Thus, designing a liquid-level sensor system that utilizes one FBG for each container is cost-effective in real environment implementation. Hence, one FBG sensor can be attached to each container and each sensor can multiplex together using a single fiber link.

To increase the number of sensors in a system and to measure

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multiple spots or containers at the same time, one important method is multiplexing the FBG sensors. Wavelength division multiplexing (WDM) is one of the most common multiplexing technique in fiber sensor system to increase the number of sensors and monitors various parameters while reducing costs [13,14]. But, one drawback of a WDM system is the limited operating range of cascaded sensors which limits the number of sensors that can be multiplexed in the sensor system. To solve this problem, multiplexing the WDM with different intensities also called intensity and wavelength division multiplexing (IWDM) method is recently been proposed [15]. IWDM can improve the FBG-based sensor system's multiplexing capability and doubles the number of sensors compared to typical WDM techniques [15].

In the IWDM, overlapping of cascaded sensors generates crosstalk, making it difficult to differentiate each sensor's central wavelength. Different techniques, such as evolution-based algorithms (EA) [16], have been applied to analyze spectra and determine the center wavelength of two gaussian overlapped spectra. However, the computation cost of evolutionary algorithms grows exponentially with the number of sensors in the system. As a result, machine learning-based methods have been adopted as a solution to this problem [17]. Beyond machine learning methods, deep learning methods have gotten attention in recent years. The development of graphics processing units (GPU) and tensor processing unit (TPU) pave the way for using a more complex algorithm such as deep neural network (DNN). Furthermore, deep learning systems need less human intervention for feature extraction and can solve more complex problems. Also, with more amount of data, deep learning algorithms are much more accurate than machine learning methods.

Accordingly, in this paper, we propose DNN to accurately predict the liquid level in multiple spots at the same time for a liquid-level sensing system. Only one FBG sensor has been used in each container to monitor the liquid level. Hence, to measure liquid levels in multiple spots at the same time, we multiplex FBG sensors that plot in a different container using IWDM. Thus, we achieve a system with lower operation and installation costs and increase the number of sensors in the liquid level sensing system. Furthermore, to increase the accuracy of water level prediction, a DNN approach is proposed rather than machine learning methods [8] have been employed. The DNN is considered to detect the central wavelength of each FBG sensor from the overlapping spectra and is used to improve the sensing accuracy. Additionally, a performance comparison is made in terms of efficacy and accuracy for the conventional machine learning methods and the proposed DNN approach. In general, the contribution of this study is summarized as follows:

- To the best of our knowledge, we propose a liquid level sensing system for measuring the water level of multiple containers at the same time for the first time, which is cost-effective.
- Since we propose a water-level sensor that only consists of one FBG sensor for each container,

which makes the sensing system is cost-effective.

- We propose DNN to predict the liquid level of each container in the liquid level sensing system, which is more accurate than conventional machine learning methods.

The rest of this paper is organized as follows. Section 2 illustrates the experimental setup and sensor structures. In section 3, the DNN model configuration is introduced, and finally, in section 4 results and comparison of this paper is discussed.

2. Experimental Setup and Sensor Structure

Fig. 1 illustrates the schematic diagram of the proposed sensor system. The system consists of four parts: (i) the central office (CO). The CO is used to generate the broadband source that emits into the FBG sensors and monitor the sensing system. The CO consists of a broadband source, optical spectrum analyzer (OSA), and optic circulator (OC). In the CO, the light generated from the broadband source is pumped into FBG sensors via 2x1 OC and transmitted into the sensing unit using a fiber transmission channel. (ii) Sensing unit: in the sensing unit, the single mood fiber (SMF) transmission channel is split into n rows by optical couplers (C). Each row has a different intensity ratio and consists of k sensors which are labeled as $S_{n,k}$ (n donate the row number and k the sensor number in the n^{th} row). Each sensor monitors the liquid level and generates a reflection spectrum dependent on the liquid level. Then, the reflected spectrum goes back to the CO through OC, and OSA is used to capture and record the reflected spectra of sensors at each stage of the liquid level. (iii) Preprocessing unit: we transfer recorded sensing data from OSA to a personal computer (PC) and then preprocessed, normalized, and prepared to feed in the DNN. (iv) Deep neural network structure: which includes the training of the proposed DNN model and predicts the liquid level of each container.

In this paper, the liquid level sensor is based on the FBG sensor. The FBGs are light-reflecting structures that transmit the broadband light spectrum while reflecting a semi-Gaussian spectrum in specific wavelengths. The Bragg wavelength (λ_B) is the center of the semi-Gaussian reflected spectrum which is the peak wavelength of the reflected spectrum of a single sensor. The Bragg wavelength can be defined by the Bragg condition [18] for a uniform structure:

$$\lambda_B = 2n_{eff}\Lambda \quad (1)$$

In Eq (1) the n_{eff} and Λ are the effective index and period of the grating, respectively. When a fiber is stressed, the center or Bragg wavelength of each FBG sensor shifts. Either compressive or tensile strain causes a change in fiber length and causes a variation in the period of the grating (Λ), which results in a shift in the Bragg wavelength. The following equation describes the relationship between wavelength change and strain [18].

$$\Delta\lambda_B = \lambda_B \times (1 - \rho\epsilon) \times \epsilon_m \quad (2)$$

Where $\rho\epsilon$ is the fiber's photo-elastic coefficient and $\epsilon_m = \Delta L/L$

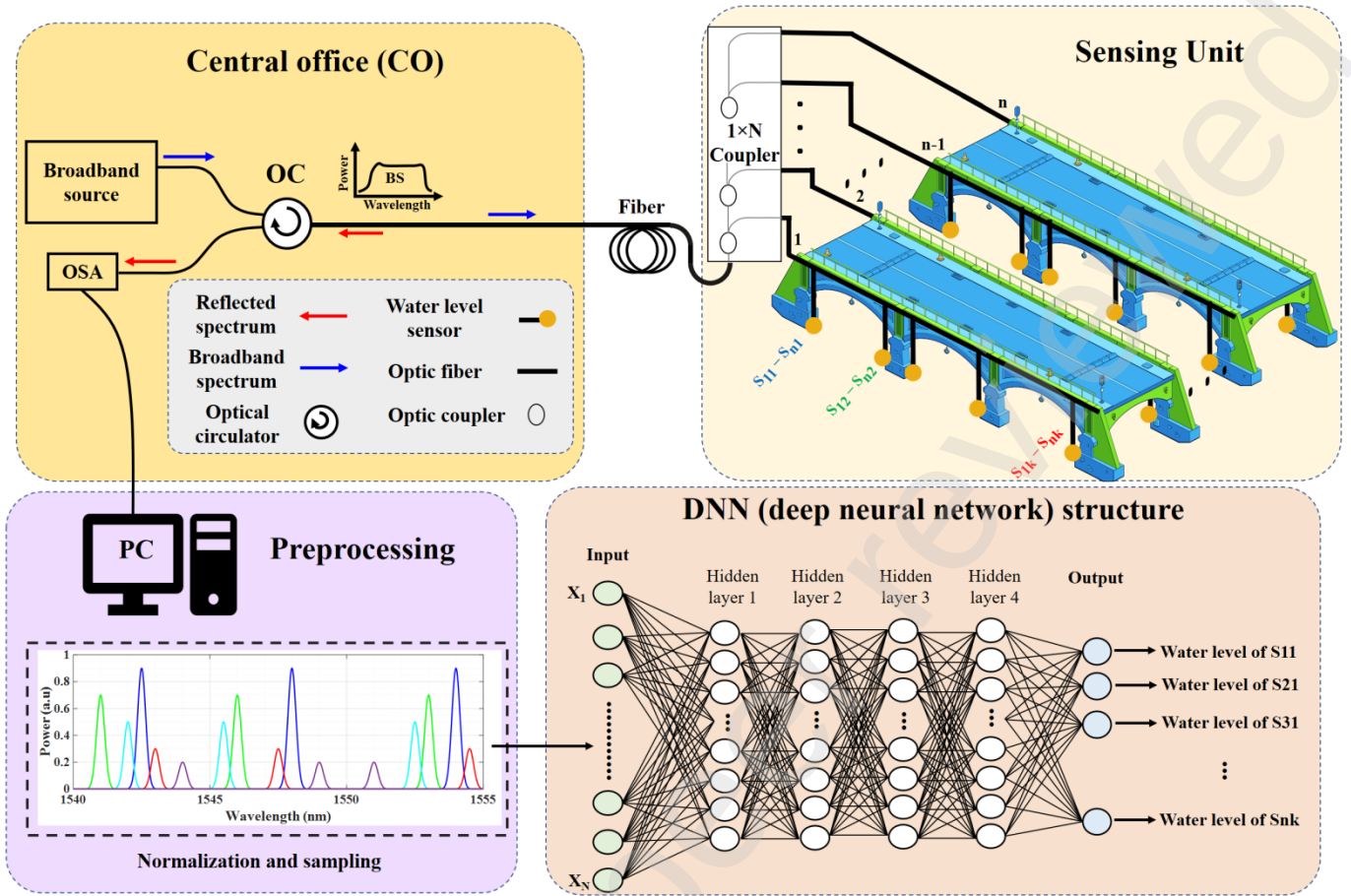


Fig. 1. Scheme diagram of the proposed multiple point liquid level sensing system which consists of four main blocks. (i) sensing unit which shows the IDWM liquid level sensing system (ii) central office (iii) preprocessing unit (iv) Deep neural network structure.

is the mechanical strain and L is the length of FBG. Due to the dependence of the center wavelength on the strain, these structures can be employed as sensors that changes in axial strain will cause the Bragg wavelength shifts. To predict the liquid level, an FBG sensor has been connected to the indicated container, as shown in Fig. 2 (a). When the water level in the container changes, the float will move upward or downward which causes axial strain and resulting a shift in the λ_B to the higher or lower wavelength.

To investigate the relation between water level changes and wavelength shift of the FBG sensor, the container has been filled with water at room temperature. Then we connected one sensor to the broadband source and apply a water level change in the container. Afterward, the OSA captured the reflected wavelength, while the water level in the container was changed by the controlled gate. As it has been shown in Fig. 2 (b) the center of the Gaussian reflected spectrum (λ_B) change from 1548.25 nm to 1549.25 nm when the water level change from 9.5 cm to 0.5 cm. Furthermore, to investigate the linearity of the sensor, we change the water level with the steps of 0.5 cm from 9.5 cm to 0.5 cm (totally 10 water level steps) and capture the reflected spectrum as well as the λ_B as shown in Fig. 2 (c). As shown in the figure, the λ_B versus water level has been shown and according to this figure, it is possible to predict the water level by a linear transformation that converts

the λ_B shifts to the water level changes.

Considering the scheme diagram of the optic fiber water level sensing system in Fig. 1, when FBGs reflection spectra are assigned to the same wavelength range, varying degrees of overlapping might occur. Even, a single peak may arise rather than several separate peaks when the reflected spectra are fully overlapping. Thus, finding the peak wavelength of each signal might be difficult. Therefore, a deep neural network (DNN) has been proposed to predict the water level according to the Bragg wavelength shift of the sensors. To train the DNN model, the reflection spectra of each FBG have been considered as $R_{i,j}$, so the total reflected spectra that are recorded in OSA can calculate as follow [19]:

$$R_{total}(\lambda) = \sum_{i=1}^n \sum_{j=1}^k R_{(i,j)} \quad (3)$$

In Eq (3), n is the number of rows and k is the number of FBG sensors in each row. The $R_{i,j}$ for each sensor can express as the following equation [17].

$$R_{i,j}(\lambda, \lambda_{B_{i,j}}) = I_{peak,i} \times \exp \left[-4 \ln 2 \times \left(\frac{\lambda - \lambda_{B_{i,j}}}{\Delta \lambda_{FWHM}} \right)^2 \right] \quad (4)$$

Where $I_{peak,i}$ is the peak reflectivity in the i^{th} row and $\Delta \lambda_{FWHM}$ is the full width at half maximum of the FBG sensor. The $\lambda_{B_{i,j}}$ is the center wavelength of FBG $_{i,j}$ that depends on its water

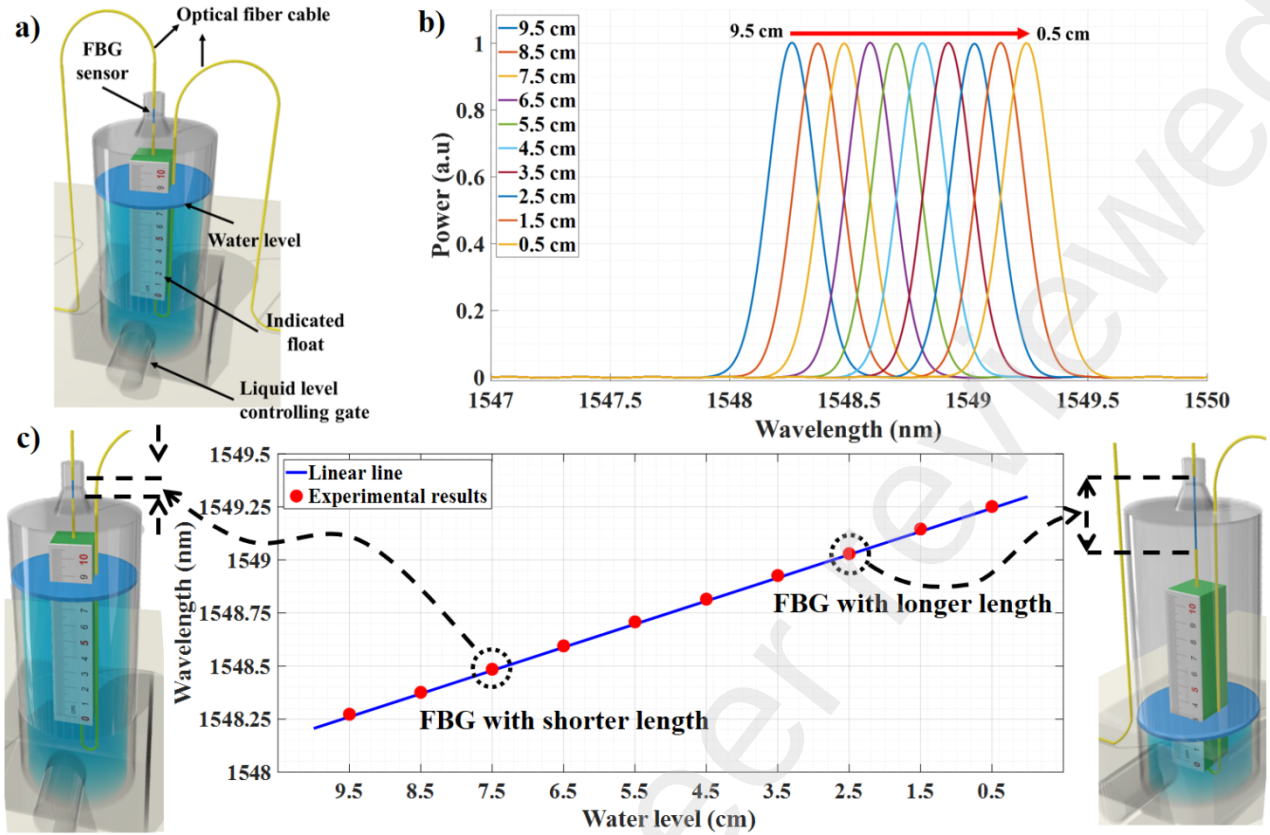


Fig. 2. (a) The water level sensor structure that an FBG sensor connected to the indicated float. The water level can control by the controlling gate. (b) The power versus wavelength of a reflected spectrum of 10 different water level steps (c) the linear shift in wavelength when the water level change from 9.5 cm to 0.5 cm with the steps of 0.5cm.

level. In industrial projects the sensor network could contain numerous rows and sensors in each row, however, in this paper, we focus on two-row ($n=2$) and one sensor in each row (S11 and S21).

3. DNN Model Configuration

Separating numerous overlapping spectra can be thought of as a regression problem, and a DNN is proposed to solve this problem. Recently, the DNN has achieved cutting-edge performance in a variety of domains, including image recognition [20], the 6G communication [21], recommendation systems [22], and visual art processing [23]. A DNN model consists of three main layers: input, hidden, and output. In a fully connected DNN, each neuron is connected to all neurons in the subsequent layer. Prediction values are obtained from the output layer after the hidden layers have processed the data features provided in the input layer. Each hidden-layer neuron has an activation function that takes the weighted sum of nodes as input and transforms it into valid values. This activation function is used to calculate predicted values and to derive the weighted total of neurons.

In this study, we consider the DNN which contains the l^{th} layer that each layer has $N[l]$ neuron, where $l = 1, 2, 3, 4, 5$ (four hidden layers and one output layer as shown in Fig. 1 in the DNN structure part). Since two sensors are investigated in the experimental setup, the $N[5] = 2$, which means there is two

output. In each layer, the biases represent by the $b[l]$ which is the $(1 \times N[l])$ vector. The $(i \times j)$ weight matrix for the l^{th} layer is represented by $w_{ij}[l]$ which i^{th} and j^{th} element of the matrix is the weight between i^{th} neuron in the previous layer and j^{th} neuron in the current layer. In each neuron, the activation function roughly applies the nonlinearity's effect in the hidden layers. Given that all of the DNN's weights and biases were obtained through a forward-propagation (FP) training, the total composite relationship between input and output is represented by Eq (5) [24].

$$Y = f(f(f(f(X^T \times w[1] + b[1]) \times w[2] + b[2]) \times w[3] + b[3]) \times w[4] + b[4]) \times w[5] + b[5]) \quad (5)$$

where, Y is a 2×1 vector ($[y_1, y_2]^T$), $f(\cdot)$ is the activation function, and $w[\cdot]$ and $b[\cdot]$ are the weight matrix and bias of each layer, respectively.

Here in, the exponential linear unit (ELU) [25] activation function has been considered for all layers, which define in Eq (6).

$$ELU(x) = \max(0, x) + \min(0, \alpha(e^x - 1)) \quad (6)$$

The α is the ELU hyperparameter and determines the point at which a negative net input ELU saturates. This activation function has been chosen based on two reasons. Firstly, it has been observed that in the hidden layers the negative values will be produced as the weights updated, although in the input layer all values are positive. Therefore, to prevent data loss ELU is a superior choice. Secondly, since we encounter a regression

problem it is recommended to use activation functions such as ReLU, PReLU, SELU, and ELU for the output layer which in this case ELU activation function shows better preference.

During each training session, the output layer forecast values and repeatedly modify the weights. Back-propagation (BP) is used in this modification of the weights from the output layer to the input layer until the loss function reaches its minimum value. According to the type of the problem, different loss functions have been suggested [26]. In the regression circumstances where it is expected that the predicted outputs are real-number values, mean squared error (MSE) is a common loss function which define as Eq (7) [27].

$$Loss = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y}_i)^2 \quad (7)$$

Where, n is the number of output-layer neurons, y_i and $\bar{y}_i$ stand for the i^{th} output's real value and i^{th} output's predicted values, respectively. Afterward, the weight is modified according to Eq (8) which the next weight is equal to the difference between the previous weight and the partial derivative of the loss function [23].

$$w_{ij}^{t+1} = w_{ij}^t - \eta \frac{\partial Loss}{\partial w_{ij}^t} \quad (8)$$

Where the w_{ij}^t donates the weight between i^{th} neuron in the previous layer and j^{th} neuron in the next layer, t is the t^{th} iteration, and η is the learning rate.

Although the DNN model has been used to predict the liquid level, it is crucial to predict the $\lambda B_{i,j}$ first. Afterward as illustrated in Fig. 2 (c), it is possible to convert $\lambda B_{i,j}$ to liquid level by a linear transform. Thus, to predict the liquid level first of all the center wavelength reflection should predict. To collect the dataset for training the DNN model and then to predict the liquid level, the OSA spectrum has been generated through the theoretical method according to Eq (3) and (4). It is recommended to train the proposed DNN model by using sensor data having distinct intensity and varied reflection spectra to improve the model learning capacity. Furthermore, it helps increase measuring accuracy as well as the versatility and adaptability of the proposed model. Therefore, we consider different peak power of each FBG sensor, respectively, and $\Delta\lambda_{FWHM} = 0.25 \text{ nm}$ for all sensors. Another important parameter for generating the dataset is the Bragg wavelength of sensors ($\lambda B_{1,1}$ and $\lambda B_{2,1}$), which has been considered 50 different λB for each sensor in its operation bandwidth (1547nm to 1550nm). As a result, 12500 different spectrum samples have been produced and 100 samples have been separated for testing, then from the remaining samples, 10% have been chosen for validation. The training dataset (i.e. captured reflected spectrum) is preprocessed and normalized between 1 to 0.

Regarding that the DNN input is the reflected spectrum, it is possible to consider different numbers of neurons for the input layer according to the number of sample points. Usually, the number of sample points is limited by the instruments' sampling accuracy. Although it is possible to increase the sample points by different interpolation methods, in some cases it reduces intentionally to reduce the system complexity and computation time. According to this trade-off between accuracy and system complexity, we consider 1001 neurons for the input

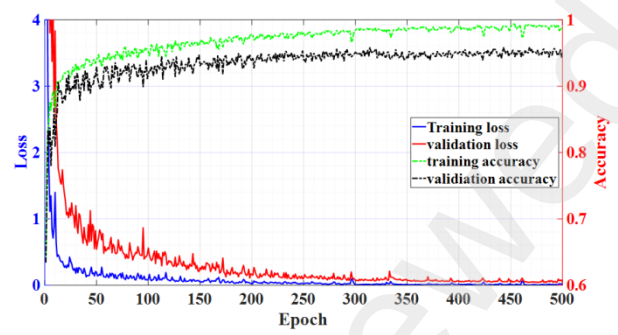


Fig. 3. The DNN model performance in terms of loss and accuracy for training and validation dataset.

layer which bring us adequate accuracy with appropriate complexity. Resultingly, the OSA simulated spectrum ($R_{total}(\lambda)$) was sampled with 1001 points of wavelength (λ). Whereas our broadband source operation bandwidth is 1547nm to 1550nm, the first wavelength sample will be $\lambda_1 = 1547\text{nm}$ and the last wavelength sample will be $\lambda_{1001} = 1550\text{nm}$ by using a wavelength shift ($\Delta\lambda$) of 0.003nm at each step. On the other hand, the number of neurons in the output layer depends on the number of sensors in the sensing system which in our case is two and labeled as the Bragg wavelength of sensors ($\lambda B_{1,1}$ and $\lambda B_{2,1}$) in the training data set. Thereby, each dataset consists of 1001 input features and 2 outputs denoted as $\{[R_{total}(\lambda_1), \dots, R_{total}(\lambda_{1001})], [\lambda B_{1,1}, \lambda B_{2,1}]\}$.

The DNN performance for the training and validation dataset is plotted in Fig. 3. In Fig. 3, the blue and red solid lines show the loss on the left axis for training and validation data, respectively, while the dashed green and black lines show the accuracy on the right axis. As shown in the figure, the training loss and validation loss converge at epoch 500 and it achieves a good performance at this epoch number. The optimal DNN parameters, including the learning factor, learning algorithm, and the number of neurons, are shown in Table 1.

Table 1: The deep neural network (DNN) parameters.

DNN parameters	value
Number of neurons in the input layer	1001
Number of hidden layers	4
Number of neurons in each hidden layer	350
Activation function	ELU
ELU hyperparameter (α)	1
Optimizer algorithm	Adam
Learning rate (η)	0.001
Batch size	250
Number of epochs	500

4. Result and Discussion

The simplified experimental setup of the proposed liquid level sensing system which has been shown in Fig. 1 is carried out by using two liquid level sensors S11 and S21 with the same operation wavelength to investigate the overlapping situation

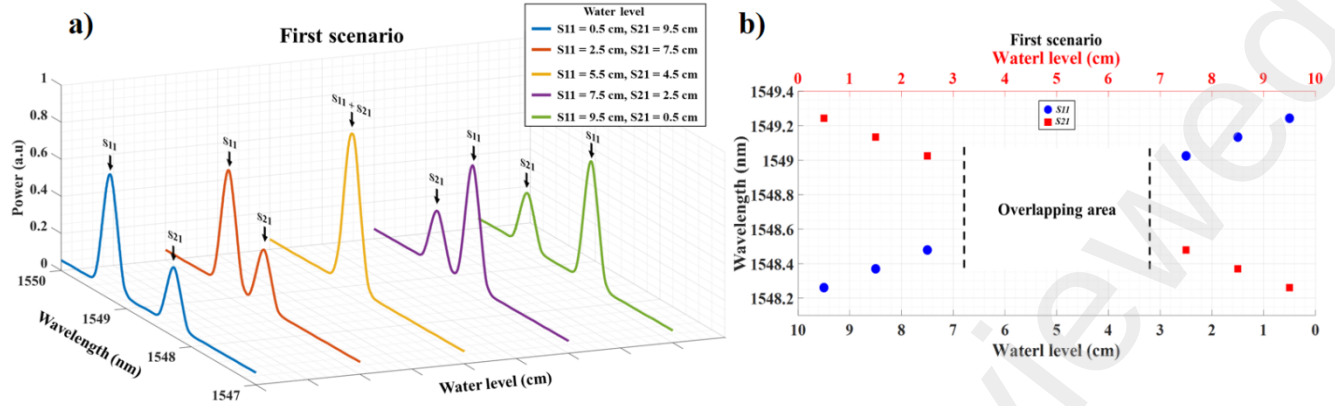


Fig. 4. (a) Reflected spectrum of two water-level sensors in 5 different water-level steps of the first scenario (b) the unmeasurable overlapping gap due to the overlap of two sensor spectra in first scenario.

and evaluate the performance of the proposed model on the experimental data. To prove the reliability and stability of our proposed deep learning algorithm, two scenarios have been considered. The first scenario is when both sensors' liquid levels change simultaneously (S11 liquid level changes from 9.5 cm to 0.5 cm and S21 liquid level changes from 0.5 cm to 9.5 cm). While in the second scenario the first sensor (S11) liquid level changes from 0.5 cm to 9.5 cm and the second one (S21) liquid level is fixed and set to be 5 cm. We fill both sensors with water at room temperature for both experiments.

4.1 First scenario: When both S11 and S21 sensors' liquid levels change simultaneously

In this scenario to change the water level of sensors simultaneously, the sensors' liquid level controlling gates are connected by a pipe. Therefore, by increasing the height of one sensor, the water flows to the other one. To change the height of the sensors, we use blocks with the height of 2 cm. It should be noted that adding a block with a height of 2 cm to a sensor, will cause the water level to reduce by 1 cm in the sensor while in another one water level will increase by 1 cm. In this scenario, 10 different water-level steps have been done, in

which the liquid level changed by 1 cm in each step. For each water-level step, we allow the water to stabilize before measuring the reflected spectrum, then the reflected spectrum have been captured using OSA. We set the OSA sampling resolution and number of sampling points to be 3 pm and 1001, respectively.

The reflected spectrum for five different water-level steps for the first scenario when the water level in sensor S11 is 0.5 cm, 2.5 cm, 5.5 cm, 7.5 cm, and 9.5 cm have been shown in Fig. 4 (a). In this figure, the fully overlapping spectra can be observed in the third plot, which is impossible to measure the sensing signals of each sensor. Fig. 4 (b) depict the water level of S11 and S21 versus their Bragg wavelength when the water level rises from 0.5 cm to 9.5 cm in the S21 but falls from 9.5 cm to 0.5 cm in the S11, which shifts in a different direction. The blue circles show the Bragg wavelength of S11 and the red squares are the Bragg wavelength of S21, for 10 different water-level steps. As shown in the figure, the unmeasurable gap (overlapping area) has occurred between S11 and S21 sensors when the water level of S11 sensor is in the range of 6.5 cm - 3.5 cm. As far as, in this scenario the Bragg wavelength of S11 and S21 shift in the opposite direction, and the overlapping area is narrow.

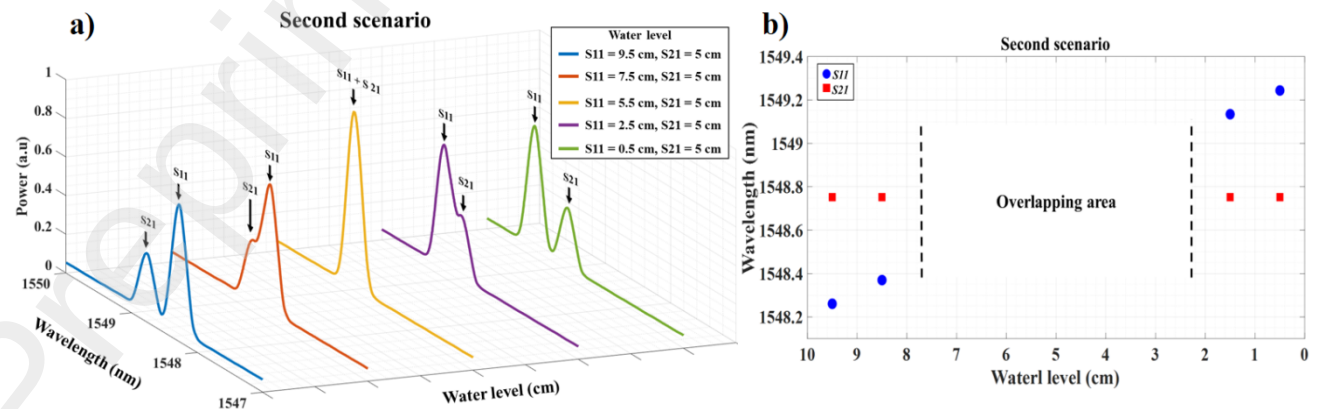


Fig. 5. (a) Reflected spectrum of two water-level sensors in 5 different water-level steps of the second scenario (b) the unmeasurable overlapping gap due to the overlap of two sensor spectra in second scenario.

4.2 Second scenario: when only S11 sensor liquid level changes

In this scenario, the first sensor's (S11) controlling gate is connected to a tank while the second sensor (S21) controlling gate has been closed. Then the S11 and S21 sensors have been filled with water to reach the 9.5 cm and 5 cm of water levels, respectively. To reduce the water level in the S11 similar method as mentioned in the first scenario is applied. Hence, the water level reduces from 9.5 cm to 0.5 cm by the steps of 1 cm in the S11, while during all these water-level steps the S21 water level remained at 5 cm. A similar setting for OSA as the first scenario has been considered.

To compare this scenario with the former one the reflected spectrum for five different water-level steps has been depicted in Fig. 5 (a) with the same amount of water in the S11 (the water level in S₁₁ is 9.5 cm, 7.5 cm, 5.5 cm, 2.5 cm, and 0.5 cm). The first, second, and third plots in this diagram show the non-overlapping, partially overlapping, and entirely overlapping regions, respectively. As it has been shown in Fig. 5 (a) in this scenario it is much more difficult to measure the sensing signals of each sensor when the reflection spectra of two sensors are partially overlapping or completely overlapping. Fig. 5 (b) shows the water level for S11 and S21 versus their Bragg wavelength when the water level in the S11 decreases from 9.5 cm to 0.5 cm and in the S21 water level is 5 cm. The S11 and S21 Bragg wavelength for 10 different water-level steps is shown with blue circles and red squares, respectively. As shown in Fig. 5 (b), the unmeasurable gap (overlapping area) has occurred between S11 and S12 sensors when the water level of S11 sensor is in the range between 7.5 cm and 2.5 cm.

In this scenario, two important reasons caused the overlapping area bigger than the former one. Firstly, the constant Bragg wavelength of S21 in all experiments, and

secondly its position, which is in the middle of sensors operation bandwidth. Consequently, predicting the water level in this scenario is more challenging rather than in the first scenario.

4.3 Performance evaluation

The performance of our proposed deep neural network model for the IWDM water-level sensing system is discussed in this section. Therefore, we have used the unseen test data of experimental results from the first and second scenarios, to test the well-trained DNN model. The test data consists of 10 water-level steps for each scenario. In the preprocessing part, the captured reflected spectrum normalized between 1 to 0. It should be noted since in the OSA setting the number of samples points have been considered to be 1001, there is no need for up/down sampling. Hence the experimental test data after normalization can be directly fed to the DNN.

The performance of the proposed model for the first and second scenarios has been shown in Fig. 6 by plotting the predicted water level versus the actual one. For both scenarios, the blue and red circles show the S11 and S21 predicted water levels respectively.

To evaluate the performance of our DNN model on the experimental test data, we use the mean absolute error (MAE) evaluation method which is defined as follows:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \bar{y}| \quad (9)$$

Where the y_i and $\bar{y}$ is the actual water level and the predicted water level, respectively, n is the number of predictions for each scenario as far as we have two sensors.

In Fig. 7 the water-level prediction error of both sensors versus water level, for each scenario has been depicted. The blue and red colors show the error for the S11 and S21,

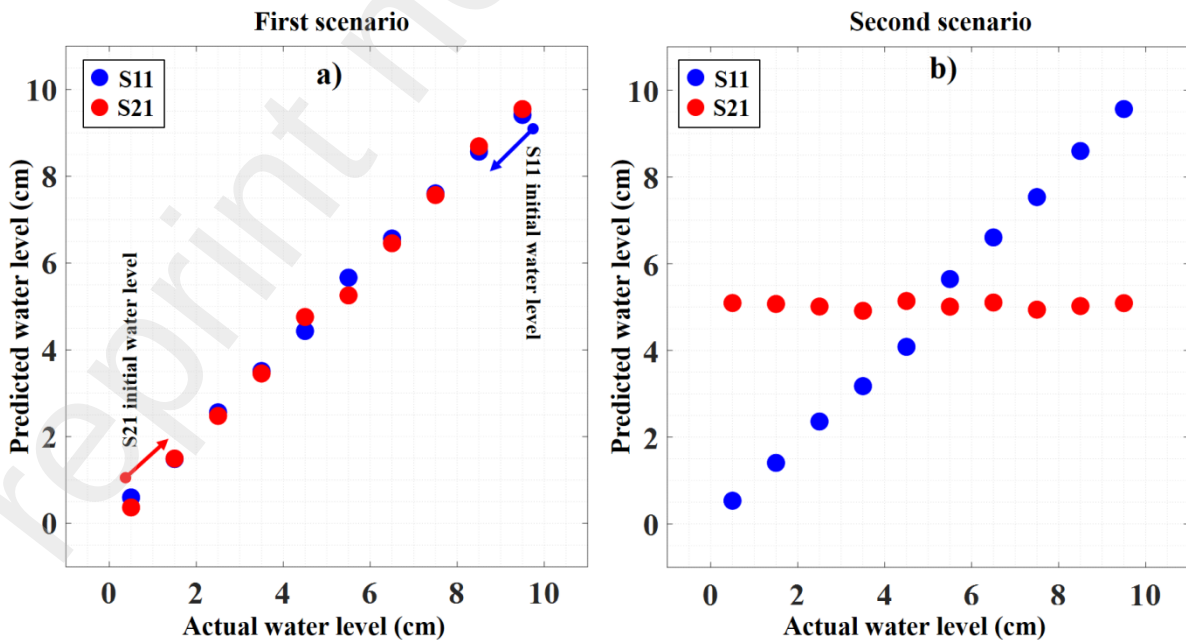


Fig. 6. The predicted water-level versus actual water-level for (a) the First scenario (b) for the second scenario.

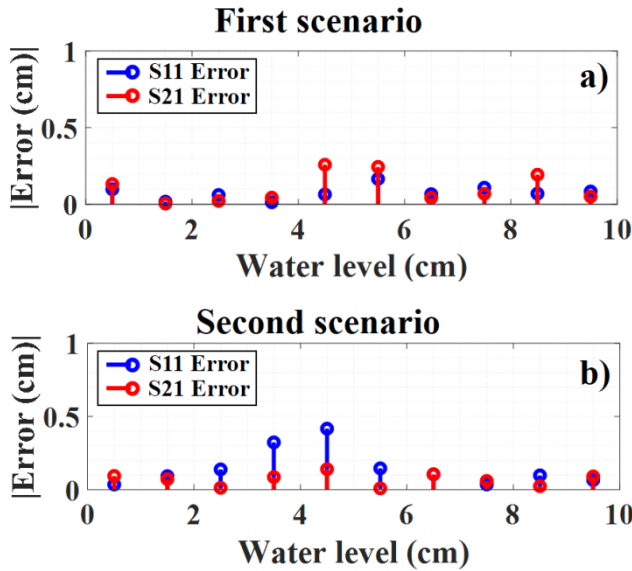


Fig. 7. The water-level prediction error versus water level for (a) the first scenario and (b) the second scenario.

respectively. As mentioned before, the bigger overlapping area in the second scenario makes prediction more challenging, thus the error in the second scenario increases slightly.

The MAE for the S11 and S21 sensors for the first scenario is 0.0748 cm and 0.1063 cm, respectively. Hence, the total MAE for the first scenario can be calculated by averaging sensors error which for the first scenario is 0.0906 cm. For the second scenario, the MAE for the S11 and S21 for the first scenario is 0.1405 cm and 0.0693 cm, and the total MAE for this scenario is 0.1074 cm. Based on the result of these two scenarios, our proposed model can predict the water-level of each sensor even from the completely overlapping spectra. Furthermore, we compare and contrast our proposed DNN approach with two conventional machine learning methods such as random forest (RF) and support vector machines (SVM) to prove the reliability of our proposed model for water-level prediction. We computed the training using the same training and testing data and under the same PC environment. To train the random forest (RF) and support vector machines (SVM), the random

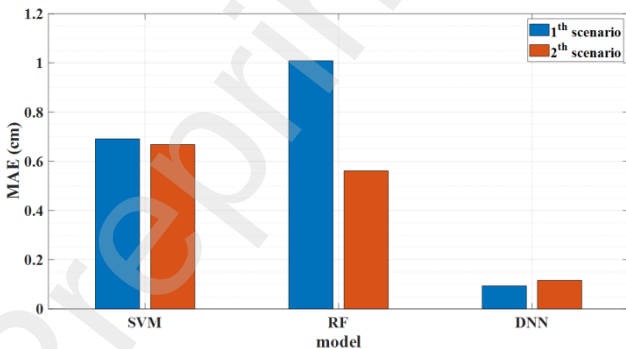


Fig. 8. Depict the mean absolute error (MAE) of experimental results for support vector machine (SVM), random forest (RF), and deep neural network (DNN) algorithm.

hyperparameter grid method is considered. Table 2 and Table 3 show the optimal parameters for training the random forest and SVM models, respectively.

Table 2: The random forest (RF) model parameters

Random forest parameters	value
The number of trees in the forest	502
The maximum depth of the tree	None
The minimum number of samples required to split an internal node	3
The minimum number of samples required to be at a leaf node	1

Table 3: The Support-vector machines (SVM) model parameters

SVM parameters	value
the kernel type	'rbf'
epsilon	0.0001
Kernel coefficient for 'rbf' (gamma)	0.01
Tolerance for stopping criterion	0.0001

Fig. 8 shows the water-level prediction performance of SVM, RF, and DNN in terms of MAE for both scenarios. The MAE of water level prediction for water level prediction based on 10 experiments which have been done in section 4 manifest the proposed DNN structure's superior performance for both scenarios, also it is worth noting that training time for DNN is much shorter than conventional machine learning method.

Additionally, to affirm the competence and capability of the proposed system, a simulation of the system with 5 sensors have been done. All five sensors (S11–S51) are considered to work in the same operating wavelength range (1547nm to 1550nm). The water level of S11 changed from 0 to 34 cm while the water level of other sensors (S21–S51) is set to be 5.7, 12.5, 19.4, and 26.2 cm, respectively. Thus, the reflected spectra of five FBGs (i.e. training dataset) at different water-level steps are collected by simulation using equation (4). The computer-simulated spectrum data of five sensors are displayed in Fig. 9 (a). As shown in the figure, it can be seen some of the spectra of two FBGs are partially or fully overlapped. Then, the generated training data can be preprocessed and prepared to be used for training DNN. Hence, the DNN model is trained with the dataset of five sensors' reflected spectrum. After the DNN model is well-trained, to prove the performance of the well-trained model, we use unseen testing data of 5 sensors' reflected spectra. As it has been depicted in Fig. 9 (b), even as the number of sensors increases, the proposed DNN model can reliably measure the water level of each container. This indicates that we can accurately predict the Bragg wavelengths of each sensor and the water level of each container even though the reflected spectra of two or more FBG sensors are partially or fully overlapped.

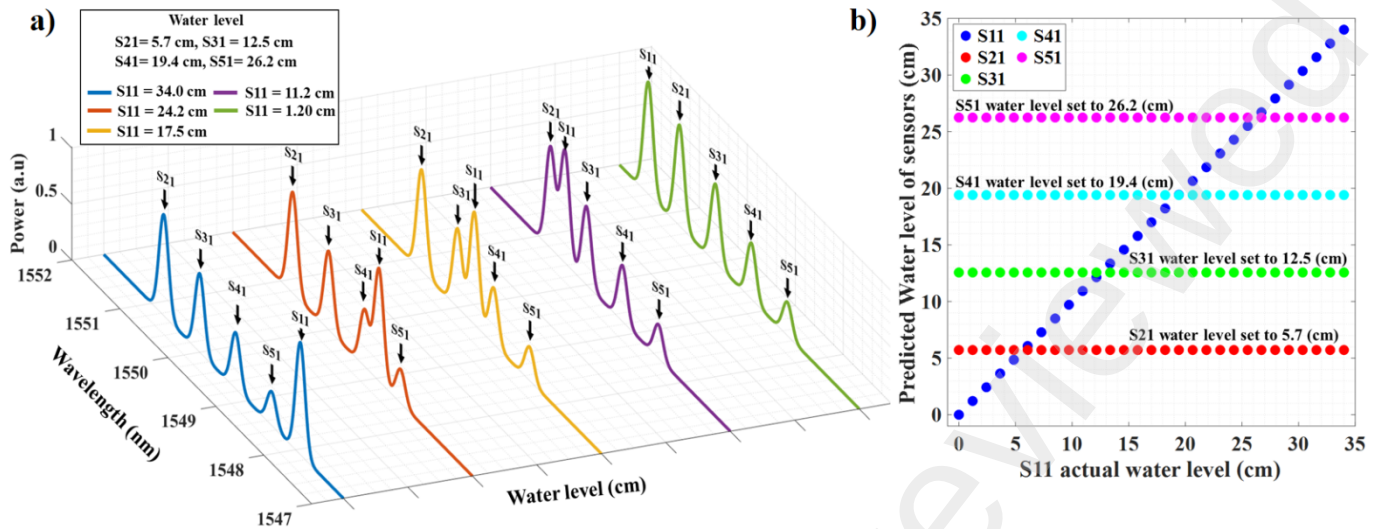


Fig. 9. (a) The reflected spectra of five water-level sensor in 5 different water-level steps (b) the predicted water level versus actual water level of five sensors.

5. Conclusion

In this paper, a novel liquid level sensing system is proposed to enhance the capacity of the sensing system and reduce the cost. The proposed sensing system can monitor the liquid level of several points at the same time which is cost-effective. Moreover, to overcome the overlapping challenge in IWDM based sensor networks, we propose a deep neural network (DNN) structure. The proposed model first predicts the Bragg wavelength and then predicts the water level. Two different scenarios have been considered to investigate the DNN performance. Hence, the mean absolute error has been used as a metrics to evaluate the DNN model water-level prediction performance. The result proves that the proposed DNN model accurately predicts the liquid level of each sensor. In the end, when compared our proposed DNN model with other conventional machine learning algorithms such as random forest (RF) and support vector machines (SVM), that the proposed DNN model exhibits the best performance.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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